IoT Challenge #3 – Part 2

LoRaWAN Exercise Report

Mobin Khatib, Ali Edareh Heidarabdi
11114435, 11036028

1 Problem Statement

The considered LoRaWAN network operates in Europe at carrier frequency 868 MHz with bandwidth 125 kHz. The network is composed of one gateway and 40 sensor nodes. Each node transmits packets with payload size 20 bytes according to a Poisson process with intensity

$$\lambda_n = 2 \text{ packets/minute.}$$

The exercise asks to determine the largest LoRa spreading factor that guarantees a packet success rate of at least 75%, assuming that all nodes use the same spreading factor and that packet transmissions are uncoordinated and may overlap in time.

2 EQ1 – Largest Spreading Factor with Packet Success Rate at Least 75%

2.1 Aggregate Traffic Rate

Since each of the $N = 40$ nodes generates packets according to a Poisson process with rate $\lambda_n = 2$ packets/minute, the total aggregate traffic generation rate is

$$\lambda_{\text{tot}} = N\lambda_n.$$

Therefore,

$$\lambda_{\text{tot}} = 40 \cdot 2 = 80 \text{ packets/minute.}$$

Converting this value into packets per second:

$$\lambda_{\text{tot}} = \frac{80}{60} \approx 1.3333 \text{ packets/second.}$$

2.2 Packet Airtime

Using the TTN airtime calculator for a 20 byte payload, EU868 region, bandwidth 125 kHz, and coding rate 4/5, the following airtime values are obtained:

As expected, the airtime increases significantly when the spreading factor increases. This happens because a higher spreading factor makes each packet stay on air for a longer time.

2.3 Pure-ALOHA Success Probability

Under the simplified pure-ALOHA model, the packet success probability is

$$P_s = e^{-2G},$$

where

Spreading Factor	Airtime [ms]	Airtime [s]
SF7	71.9	0.0719
SF8	133.6	0.1336
SF9	246.8	0.2468
SF10	493.6	0.4936
SF11	987.1	0.9871
SF12	1974.3	1.9743

Table 1: Packet airtime values obtained from the TTN airtime calculator.

$$G = \lambda_{\text{tot}} T.$$

Here, T is the packet airtime and G is the normalized offered load. Therefore,

$$P_s = e^{-2\lambda_{\text{tot}} T}.$$

The required condition is

$$P_s \geq 0.75.$$

Substituting the pure-ALOHA expression:

$$e^{-2\lambda_{\text{tot}} T} \geq 0.75.$$

Taking the natural logarithm:

$$-2\lambda_{\text{tot}} T \geq \ln(0.75).$$

Thus, the maximum allowed airtime is

$$T_{\text{max}} = \frac{-\ln(0.75)}{2\lambda_{\text{tot}}}.$$

Substituting $\lambda_{\text{tot}} = 1.3333$ packets/s:

$$T_{\text{max}} = \frac{-\ln(0.75)}{2 \cdot 1.3333}.$$

$$T_{\text{max}} \approx 0.1079 \text{ s}.$$

Therefore,

$$T_{\text{max}} \approx 107.9 \text{ ms}.$$

Any spreading factor whose airtime is larger than 107.9 ms cannot satisfy the 75% success-rate requirement.

2.4 Numerical Evaluation

The success probability is computed for each spreading factor using

$$P_s = e^{-2\lambda_{\text{tot}} T}.$$

For SF7:

$$G_{\text{SF7}} = 1.3333 \cdot 0.0719 \approx 0.0959,$$

$$P_{s,\mathrm{SF7}} = e^{-2 \cdot 0.0959} \approx 0.8255.$$

Therefore,

$$P_{s,\mathrm{SF7}} \approx 82.6\%.$$

For SF8:

$$G_{\mathrm{SF8}} = 1.3333 \cdot 0.1336 \approx 0.1781,$$

$$P_{s,\mathrm{SF8}} = e^{-2 \cdot 0.1781} \approx 0.7003.$$

Therefore,

$$P_{s,\mathrm{SF8}} \approx 70.0\%.$$

For SF9:

$$G_{\mathrm{SF9}} = 1.3333 \cdot 0.2468 \approx 0.3291,$$

$$P_{s,\mathrm{SF9}} = e^{-2 \cdot 0.3291} \approx 0.5178.$$

Therefore,

$$P_{s,\mathrm{SF9}} \approx 51.8\%.$$

For SF10:

$$G_{\mathrm{SF10}} = 1.3333 \cdot 0.4936 \approx 0.6581,$$

$$P_{s,\mathrm{SF10}} = e^{-2 \cdot 0.6581} \approx 0.2682.$$

Therefore,

$$P_{s,\mathrm{SF10}} \approx 26.8\%.$$

For SF11:

$$G_{\mathrm{SF11}} = 1.3333 \cdot 0.9871 \approx 1.3161,$$

$$P_{s,\mathrm{SF11}} = e^{-2 \cdot 1.3161} \approx 0.0719.$$

Therefore,

$$P_{s,\mathrm{SF11}} \approx 7.2\%.$$

For SF12:

$$G_{\mathrm{SF12}} = 1.3333 \cdot 1.9743 \approx 2.6324,$$

$$P_{s,\mathrm{SF12}} = e^{-2 \cdot 2.6324} \approx 0.0052.$$

Therefore,

$$P_{s,\mathrm{SF12}} \approx 0.52\%.$$

The results are summarized in Table 2.

SF	T [s]	$G = \lambda_{\mathbf{tot}} T$	$P_s = e^{-2G}$	**Verdict**
SF7	0.0719	0.0959	0.8255 (82.6%)	Pass
SF8	0.1336	0.1781	0.7003 (70.0%)	Fail
SF9	0.2468	0.3291	0.5178 (51.8%)	Fail
SF10	0.4936	0.6581	0.2682 (26.8%)	Fail
SF11	0.9871	1.3161	0.0719 (7.2%)	Fail
SF12	1.9743	2.6324	0.0052 (0.52%)	Fail

Table 2: Pure-ALOHA packet success probability for different spreading factors.

2.5 Discussion

From the results, only SF7 satisfies the required success-rate condition. SF8 already fails because its airtime is 133.6 ms, which is larger than the maximum allowed airtime of 107.9 ms. For SF9, SF10, SF11, and SF12, the packet airtime becomes even longer, so the offered load G increases and the packet success probability decreases further.

This shows the main trade-off of LoRa spreading factor selection. A higher spreading factor improves receiver sensitivity and can increase communication range, but it also increases packet airtime. In a collision-limited pure-ALOHA model, longer airtime increases the probability of packet overlap, reducing the packet success rate.

2.6 Answer to EQ1

From the airtime threshold,

$$T_{\max} \approx 107.9 \text{ ms},$$

only SF7 satisfies the 75% success-rate requirement. Therefore, the largest spreading factor that guarantees a packet success rate of at least 75% is

$$\boxed{\text{SF7}}.$$

The predicted success rate for SF7 is

$$\boxed{P_{s,\text{SF7}} \approx 82.6\%}.$$

3 EQ2 – Diagnosing Non-Uniform Field Performance

After deploying the network in the field, the measured packet success rate is significantly lower than expected. Moreover, the performance is not uniform across nodes: some nodes achieve relatively good delivery rates, while others experience very poor performance.

The proposed actions are:

1. Decrease the number of nodes.
2. Move the nodes closer to the gateway.
3. Increase the spreading factor.

3.1 Interpretation of the Field Observation

The pure-ALOHA model used in EQ1 considers only collision losses due to overlapping transmissions. In this model, losses are mainly determined by the aggregate traffic load and packet airtime. Therefore, if collisions were the only dominant problem, the degradation would be expected to affect the nodes in a more uniform way.

However, the field observation says that performance is not uniform: some nodes work well, while others perform very poorly. This suggests that the problem is not only due to collisions, but also due to link-budget differences between nodes.

In a real LoRaWAN deployment, the received power depends on distance from the gateway, obstacles, antenna orientation, fading, and propagation conditions. Some nodes may be close to the gateway or have a clear propagation path, while others may be farther away or blocked by obstacles. These weaker nodes may have received power close to or below the receiver sensitivity threshold, causing poor packet delivery.

3.2 Evaluation of the Proposed Actions

1. Decrease the number of nodes. Reducing the number of nodes would reduce the aggregate traffic rate λ_{tot} and therefore reduce the collision probability. This could improve the overall packet success rate if collisions were the main problem. However, it does not directly solve the non-uniform performance among nodes. The nodes with weak radio links would still have poor link quality.

2. Move the nodes closer to the gateway. Moving the nodes closer to the gateway directly improves the link budget. The received power increases because the path loss is reduced. This action is especially useful for the nodes that currently have poor delivery rates, because they are likely the nodes suffering from weak signal conditions.

Moreover, this solution does not increase the packet airtime. Therefore, it improves the physical-layer link quality without increasing the collision probability.

3. Increase the spreading factor. Increasing the spreading factor improves receiver sensitivity and can help weak links. However, a higher spreading factor also increases packet airtime. A longer airtime increases the probability of packet overlap and collisions.

From EQ1, increasing SF from SF7 to SF8 already decreases the predicted success rate from about 82.6% to about 70.0%. Higher spreading factors reduce the success probability even more: SF9 gives about 51.8%, SF10 gives about 26.8%, SF11 gives about 7.2%, and SF12 gives only about 0.52%. Therefore, increasing the spreading factor would improve coverage but would also strongly worsen the collision-limited performance of the network.

3.3 Answer to EQ2

The most appropriate action is

$$\boxed{\text{2. Move the nodes closer to the gateway.}}$$

The reason is that the non-uniform performance across nodes indicates a link-budget problem rather than only a collision problem. Moving the nodes closer to the gateway improves the radio link quality for the poorly performing nodes without increasing the airtime and without increasing the collision probability.

4 Final Answers

- **EQ1:** The largest spreading factor that guarantees a packet success rate of at least 75% is

  $$\boxed{\text{SF7}}$$

  with predicted packet success rate

  $$\boxed{P_s \approx 82.6\%}.$$

- **EQ2:** The most appropriate action is

  $$\boxed{\text{Move the nodes closer to the gateway.}}$$

  This directly improves the link budget of the poorly performing nodes without increasing packet airtime or collision probability.